

CS3700 — In-Class Activity: Membership Records

This activity requires you to **work in pairs**.

A gym maintains two separate sets of records:

- **StandardMember:** MemberID, Name, Email, JoinDate
- **PremiumMember:** MemberID, Name, Email, JoinDate, MonthlyTrainingHours

The gym wants to combine these into one model. MemberID uniquely identifies a person across both sets of records. Every member has exactly one membership type: standard or premium.

Your Tasks

1. Would you use **generalization or specialization** to reorganize these existing records? Explain briefly.
2. Identify the superclass, subclasses, and shared attributes.
3. Draw an **ERD using Chen notation**
4. Is the hierarchy **disjoint or overlapping**? **Total or partial**? Justify your answers.

Challenge: The gym starts offering day passes. Day-pass members have only the shared attributes and are neither standard nor premium members. How would this affect the existing hierarchy?